English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of all eleven handwritten pages. Calculations are preserved as written; this is a translation, not a corrected solution set.

Problem 1

The density is supported on $x, y \geq 0$, $y \leq (1-x)/2$, equivalently $x \leq 1 - 2y$.

1. Normalize:

$$\begin{aligned} 1 &= \int_0^1 \int_0^{(1-x)/2} (cx^2+1)\,dy\,dx = \frac{1}{2}\int_0^1 (cx^2+1)(1-x)\,dx \\ &= \frac{1}{2}\int_0^1 (cx^2+1-cx^3-x)\,dx = \frac{1}{2}\left[\frac{c}{3}+1-\frac{c}{4}-\frac{1}{2}\right] = \frac{c}{24}+\frac{1}{4}. \end{aligned}$$

Hence $\boxed{c = \frac{3}{4}\cdot 24 = 18}$. For $x \in (0,1)$,

$$\boxed{f_X(x) = \int_0^{(1-x)/2} (cx^2+1)\,dy = (cx^2+1)\frac{1-x}{2}.}$$

Furthermore,

$$\begin{aligned} \mathbb{E}Y &= \int_0^1 \int_0^{(1-x)/2} y(cx^2+1)\,dy\,dx \\ &= \int_0^1 (cx^2+1)\left(\frac{1-x}{2}\right)^2 \frac{1}{2}\,dx \\ &= \frac{1}{8}\int_0^1 (cx^2+1)(1-2x+x^2)\,dx \\ &= \frac{1}{8}\int_0^1 (cx^2+1-2cx^3-2x+cx^4+x^2)\,dx \\ &= \frac{1}{8}\left[6+1-9-1+\frac{18}{5}+\frac{1}{3}\right] = \frac{1}{8}\frac{54+5-45}{15} = \frac{14}{120} = \boxed{\frac{7}{60}}. \end{aligned}$$

2. $$f_{Y|X}(y|\tfrac{1}{2}) = \frac{f_{X,Y}(\frac{1}{2},y)}{f_X(\frac{1}{2})} = \frac{c(\frac{1}{2})^2+1}{(c(\frac{1}{2})^2+1)\frac{1-\frac{1}{2}}{2}} = 4, \qquad 0 < y < \tfrac{1}{4}.$$

Thus $\mathbb{E}[Y|X=1/2] = \mathbb{E}[U(0,1/4)] = \boxed{1/8}$.

3. For $t \in (0,1)$,

$$\begin{aligned} \mathbb{P}(X+2Y \leq t) = \mathbb{P}(Y \leq (t-X)/2) &= \int_0^{t/2}\int_0^{t-2y} (cx^2+1)\,dx\,dy \\ &= \int_0^{t/2}\left[c\frac{(t-2y)^3}{3}+t-2y\right]dy \\ &= \frac{1}{2}\int_0^t \left(\frac{cw^3}{3}+w\right)dw = \frac{1}{2}\left[\frac{18t^4}{12}+\frac{t^2}{2}\right] = \frac{1}{4}(3t^4+t^2). \end{aligned}$$

The substitution is $w = t - 2y$. Hence

$$\boxed{f_Z(t) = 3t^3 + \tfrac{1}{2}t, \quad 0 < t < 1; \qquad f_Z(t) = 0 \text{ otherwise.}}$$

Alternatively use $Z = X + 2Y$, $W = X$, whose inverse is $X = W$, $Y = (Z-W)/2$. Its absolute Jacobian is

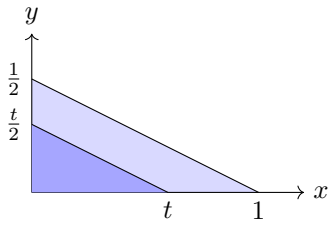
$$|J| = \begin{vmatrix} 1 & 0 \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2}.$$

Then

$$f_{Z,W}(z,w) = f_{X,Y}(w,(z-w)/2)\frac{1}{2} = (cw^2+1)\frac{1}{2}$$

on the transformed region, namely $z \in (0,1)$, $w \in (0,z)$. Thus

$$f_Z(t) = \int_0^t (cw^2+1)\frac{1}{2}\,dw = 3t^3 + \frac{t}{2}, \qquad 0 < t < 1.$$



Problem 2

Let $X_1, X_2, \cdots \sim \text{Exp}(1/5)$ be the times between customers under 30, and $Y_1, Y_2, \cdots \sim \text{Exp}(1/10)$ those between customers over 30. We have $\mathbb{E}X_i = 5$, $DX_i = 25$.

1.

$$\mathbb{P}(X_1 + \cdots + X_{100} < 480) = \mathbb{P}\left(\frac{X_1 + \cdots + X_{100} - 100 \cdot 5}{\sqrt{25}\sqrt{100}} < \frac{-20}{\sqrt{25 \cdot 100}}\right) \approx \Phi(-0.4) \approx \boxed{34.458\%}.$$

2.

$$Z_1 := \min(X_1, Y_1) \sim \text{Exp}(\tfrac{1}{5} + \tfrac{1}{10}) = \text{Exp}(\tfrac{3}{10}).$$

3. If Z_i is the time between customers $i-1$ and i, then $Z_i \sim \text{Exp}(3/10)$, $\mathbb{E}Z_i = 10/3$, $DZ_i = (10/3)^2$. Hence

$$\mathbb{P}(Z_1 + \cdots + Z_{150} < 480) = \mathbb{P}\left(\frac{Z_1 + \cdots + Z_{150} - 500}{\sqrt{150}\,(10/3)} < \frac{-20}{\sqrt{150}\,(10/3)}\right) \approx \Phi(-0.49) \approx 31.287\%.$$

Both approximations use the CLT.

Problem 3

Let X_i, Y_i be the horizontal and vertical components of step i. Each of $(1,0),(0,1),(-1,0),(0,-1)$ has probability $1/4$. Thus $\mathbb{E}X_1 = \frac{1}{4} - \frac{1}{4} + 0 \cdot \frac{1}{2} = 0$ and

$$\mathbb{E}X_1^2 = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}, \qquad \boxed{DX_1 = \tfrac{1}{2}}.$$

1.

$$\mathbb{P}(X_1 + \cdots + X_{1000} > 10) = \mathbb{P}\left(\frac{X_1 + \cdots + X_{1000}}{\sqrt{1/2}\sqrt{1000}} > \frac{10}{\sqrt{500}}\right).$$

The handwritten CLT approximation is written as $\Phi(0.45) \approx \boxed{32.636\%}$.

2. Write d for distance after 100 steps. By symmetry,

$$\begin{aligned}\mathbb{E}d^2 &= \mathbb{E}[(X_1 + \cdots + X_{100})^2 + (Y_1 + \cdots + Y_{100})^2] \\ &= 2\mathbb{E}[(X_1 + \cdots + X_{100})^2] \\ &= 2\left(100\mathbb{E}X_1^2 + 2\sum_{i \neq j}\mathbb{E}[X_iX_j]\right) = 200\mathbb{E}X_1^2 = \boxed{100}.\end{aligned}$$

The mixed expectations vanish because different steps are independent and $\mathbb{E}X_i = 0$. For the bonus, use $(X_i, Y_i) \perp\!\!\!\perp (X_j, Y_j)$ when $i \neq j$, and $X_iY_i = 0$ always:

$$\begin{aligned}\mathbb{E}[(d^2)^2] = 2\mathbb{E}[(X_1 + \cdots + X_{100})^4] \\ + 2\mathbb{E}[(X_1 + \cdots + X_{100})^2(Y_1 + \cdots + Y_{100})^2]\end{aligned}$$

$$= 2\left(100\mathbb{E}X_1^4 + \binom{100}{2}\binom{4}{2}(\mathbb{E}X_1^2)^2\right)$$
$$+ 2\left(100\mathbb{E}[X_1^2Y_1^2] + 100\cdot 99(\mathbb{E}X_1^2)^2\right)$$
$$= 2\left(100\cdot\tfrac{1}{2} + 50\cdot 99\cdot 6(\tfrac{1}{2})^2\right) + 200\cdot 99(\tfrac{1}{2})^2.$$

Here $\mathbb{E}[X_1^2Y_1^2] = 0$. Therefore

$$D(d^2) = \mathbb{E}[(d^2)^2] - (\mathbb{E}d^2)^2 = \boxed{9900}.$$

For n steps, $\mathbb{E}d^2 = n$ and

$$\mathbb{E}[(d^2)^2] = 2\mathbb{E}[(X_1+\cdots+X_n)^4] + 2\mathbb{E}[(X_1+\cdots+X_n)^2(Y_1+\cdots+Y_n)^2]$$
$$= 2n\mathbb{E}X_1^4 + 2\frac{n(n-1)}{2}\,6(\mathbb{E}X_1^2)^2 + 2n(n-1)(\mathbb{E}X_1^2)^2$$
$$= n + \tfrac{3}{2}n(n-1) + \tfrac{1}{2}n(n-1) = 2n^2 - n.$$

Hence $\boxed{D(d^2) = 2n^2 - n - n^2 = n^2 - n}$.

Problem 4

1. $\mathbb{E}X_n = 2\mathbb{E}X_{n-1}$, so $\mathbb{E}X_n = 2^n$. Also $DX_n = 4DX_{n-1} + 1$, with $DX_0 = 1$. Set $a_n = DX_n$. From $a_n = 4a_{n-1} + 1$ and $a_{n-1} = 4a_{n-2} + 1$,

$$a_n - 5a_{n-1} + 4a_{n-2} = 0.$$

The characteristic equation $t^2 - 5t + 4 = 0$ has roots 4 and 1, hence $a_n = C_1\,4^n + C_2$. Since $a_0 = 1, a_1 = 5$,

$$C_1 + C_2 = 1, \quad 4C_1 + C_2 = 5, \quad C_1 = \tfrac{4}{3}, \quad C_2 = -\tfrac{1}{3}.$$

Thus $\boxed{DX_n = \tfrac{1}{3}(4^{n+1} - 1)}$.

2. $\mathbb{E}(X_i - 2X_{i-1})^2 = \mathbb{E}\epsilon_i^2 = 1$, so $\boxed{\mathbb{E}S_{n,1} = n}$. For the absolute-value term, the handwritten intermediate line is $\mathbb{E}|X_n - 2X_{n-1}|^2 = \mathbb{E}|\epsilon_i|$. The notes then use

$$\mathbb{E}|N(0,1)| = \int_{-\infty}^{\infty} |x|\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\,dx$$
$$= 2\int_0^{\infty} \frac{x}{\sqrt{2\pi}}e^{-x^2/2}\,dx = \frac{2}{\sqrt{2\pi}}\int_0^{\infty} e^{-x^2/2}\,d(x^2/2) = \sqrt{\frac{2}{\pi}}.$$

Therefore $\boxed{\mathbb{E}S_{n,2} = n\sqrt{2/\pi}}$.

3. If $\epsilon_i \sim N(1,2)$, then $\mathbb{E}\epsilon_i = 1$, $D\epsilon_i = 2$. We have $\mathbb{E}X_n = 2\mathbb{E}X_{n-1} + 1$, $\mathbb{E}X_0 = 1$. For $a_n = 2a_{n-1} + 1$, $a_0 = 1, a_1 = 3$,

$$a_n - 3a_{n-1} + 2a_{n-2} = 0, \qquad a_n = C_1\,2^n + C_2.$$

The equations $C_1 + C_2 = 1$, $2C_1 + C_2 = 3$ give $\boxed{\mathbb{E}X_n = 2^{n+1} - 1}$. Similarly, $DX_n = 4DX_{n-1} + 2$ has form $C_1\,4^n + C_2$, with $DX_0 = 1, DX_1 = 6$, giving

$$\boxed{DX_n = \tfrac{1}{3}(5\cdot 4^n - 2)}.$$

The notes calculate

$$\mathbb{E}S_{n,1} = n\mathbb{E}[N(1,2)^2] = n(2^2 - 1) = 3n,$$
$$\mathbb{E}S_{n,2} = n\mathbb{E}|N(1,2)|^2 = n\mathbb{E}|\sqrt{2}Z + 1|, \quad Z \sim N(0,1).$$

Writing φ for the standard normal density,

$$\mathbb{E}|\sqrt{2}Z + 1| = \int_{-\infty}^{\infty} |\sqrt{2}x + 1|\varphi(x)\,dx$$
$$= \int_{-\sqrt{2}/2}^{\infty} (\sqrt{2}x + 1)\varphi(x)\,dx - \int_{-\infty}^{-\sqrt{2}/2} (\sqrt{2}x + 1)\varphi(x)\,dx$$

$$
\begin{aligned}
&= \int_{-\sqrt{2}/2}^{\infty} \varphi(x)\,dx - \int_{-\infty}^{-\sqrt{2}/2} \varphi(x)\,dx \\
&\quad + \int_{-\sqrt{2}/2}^{\infty} \sqrt{2}x\varphi(x)\,dx + \int_{\sqrt{2}/2}^{\infty} \sqrt{2}x\varphi(x)\,dx \\
&= \Phi(\tfrac{\sqrt{2}}{2}) - (1 - \Phi(\tfrac{\sqrt{2}}{2})) + \frac{1}{\sqrt{\pi}} \left[\int_{-\sqrt{2}/2}^{\sqrt{2}/2} xe^{-x^2/2}\,dx + 2\int_{\sqrt{2}/2}^{\infty} xe^{-x^2/2}\,dx \right] \\
&= 2\Phi(\tfrac{\sqrt{2}}{2}) - 1 + \frac{2}{\sqrt{\pi}} e^{-1/4} \approx 1.3993.
\end{aligned}
$$

The symmetric integral is zero.

The final general formula in the handwritten source is

$$
X \sim N(\mu, \sigma^2): \qquad \mathbb{E}|X| = \mu(2\Phi(\mu/\sigma) - 1) + \sigma\sqrt{\frac{2}{\pi}} e^{-\mu^2/(2\sigma^2)}.
$$

See the folded normal distribution.